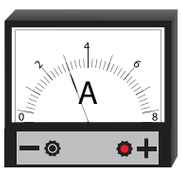
This movement of positive charges is called **conventional current.**

After the electrons were discovered, it was known that electron flow actually takes place from the negative terminal to the positive terminal of the battery. This movement is known as **electron flow.**

Conventional current is in the direction opposite to electron flow.

### 2.1.2. Measurement of electric current




Electric current is measured using a device called ammeter. The terminals of an ammeter are marked with + and - sign. An ammeter must be connected in series in a circuit.

Instruments used to measure smaller currents, in the milli ampere or micro ampere range, are designated as milli ammeters or micro ammeters.

1 milliampere (mA) = $10^{-3}$ ampere.
= 1/1000 ampere
1 microampere (μA) = $10^{-6}$ ampere
= 1/1000000 ampere

## Worked Examples 2.2

If 0.002A current flows through a circuit, then convert the current in terms of micro ampere?

### Solution:

Given that the current flows through the circuit is 0.002A

We know that

$1\ A = 10^{6}\ \mu A$

$0.002A = 0.002 \times 10^{6}\ \mu A$

$= 2 \times 10^{-3} \times 10^{6}\ \mu A$

$= 2 \times 10^{3}\ \mu A$

$0.002A = 2000\ \mu A$

### 2.2. Potential difference (V)

Electrical charges need energy to push them along a circuit.

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Water always flows from higher to lower ground. Similarly an electric charge always flows from a point at higher potential to a point at lower potential.

An electric current can flow only when there is a potential difference (V) or P.D.

The potential difference between any two points in the circuit is the amount of energy needed to move one unit of electric charge from one point to the other.

### 2.2.1. Unit of potential difference

Did you ever notice the precautionary board while crossing the railway track and the electrical transformer? What does the word high voltage denotes?





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